

PEDRO BARBOSA

Backend Engineer — Python, SQL, Distributed Tooling

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Portuguese (native) · English (fluent) · German (basic)

SUMMARY

Backend engineer working on production systems, focused on Python and data. I build services and tooling that other people depend on, and I write the automated tests that keep them correct. Strong in Python (OOP, pytest, pandas) and SQL modelling, with hands-on Docker and CI pipelines, and used to debugging live systems while real users are on them. I close knowledge gaps fast: I taught myself Go and shipped a 5,000-line daemon with more test code than production code, and I have an accepted contribution in Akaunting, an open source accounting platform.

EXPERIENCE

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform used daily by patients, clinics and professionals.
- Own features end to end: relational schema and migrations, REST endpoints, interface, and the automated tests around them.
- Wrote 23 of the repository's 50 database migrations and the largest share of its test suite.
- Handle production incidents: reproduce the failure, find the root cause, fix, and add a regression test so it cannot return silently.

Quantitative Developer — QuantConnect (freelance, remote)

September 2025 – present

- Python on the LEAN engine: backtesting pipelines over large financial time series, factor research and risk models.
- Exact numerical handling and reproducible results; batch simulations over years of historical market data.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Django system for a non-profit: role-based access control, email verification, privacy-compliant handling of personal data, normalised PostgreSQL/MySQL schema designed to stay fast as volume grows.
- Data pipelines with pandas and openpyxl replacing manual spreadsheet work, removing a recurring source of human error.
- Power BI dashboards over fleet, HR and inventory data.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend: orders, authentication, delivery tracking, payment gateway integration.
- MySQL modelling and documentation, unit tests with pytest, optimisation of slow queries and HTTP requests.

SKILLS

Languages: Python, Go, PHP, TypeScript, SQL, Bash

Python: OOP, pytest, pandas, Django, asyncio

Data: PostgreSQL, MySQL, SQLite, scikit-learn

Infrastructure: Docker, GitHub Actions, GitLab CI, Linux, systemd

Cloud and networking: working knowledge of cloud platforms and networking primitives (VPC, security groups, load balancers) from coursework and personal projects

Practices: automated testing (unit, integration), code review, versioned migrations, incident debugging, LLM-assisted development with Claude

EDUCATION

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (expected) · GPA 8.7/10

Scientific initiation in Economic Mathematics — IMPA, 2024 · Machine Learning certificate — IMPA, 2024

CERTIFICATIONS

HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python · Coursera: Python for Data Science and AI